Essays

|!|BCEEB0180000001|!| ★ **Cross topic**

When a person accidentally touches a hot object, a reflex action occurs to withdraw the hand quickly. Describe the role of the nervous system and the musculo-skeletal system in this reflex action. (12 marks)

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Stimulation by the stimulus and generation of nerve impulses (Max. 3)

- When a person touches a hot object, the receptors in the skin are stimulated to generate nerve impulses. (1)
- Nerve impulses are transmitted along the sensory neurone to the interneurone in the spinal cord. (1)
- In the spinal cord, nerve impulses are transmitted from the interneurone to the motor neurone. (1)

Initiation of muscle contraction by nerve impulses (Max. 3)

- The arrival of nerve impulses at the ending of the motor neurone stimulates the vesicles to release neurotransmitters. (1)
- Neurotransmitters diffuse across the cleft in the neuromuscular junction. (1)
- Neurotransmitters bind to receptor sites on the membrane of a muscle fibre. (1)
- Binding of neurotransmitters stimulates muscle fibres to generate electrical impulses. (1)
- Electrical impulses spread over the muscle fibres, causing muscle contraction. (1)

Actions of antagonistic muscles (Max. 3)

- The biceps and triceps in the arm are an antagonistic muscle pair. (1)
- One end of the biceps is attached to the radius by inelastic tendons. (1)
- When the biceps is stimulated to contract, it shortens and the pulling force is transmitted to the radius / forearm through the tendon. (1)
- At the same time, the triceps relaxes to allow the arm to bend (1).

Communication (Max. 3)

(12 marks)

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Skills assessed	Marks
Understanding basic concepts	9
Communication	3

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|!|BCEEB0180000002|!| ★ **Cross topic**

Compare and contrast the transmission of a nerve impulse across a synapse and a neuromuscular junction. (11 marks)

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Similarities (Max. 4)

- Both a synapse and a neuromuscular junction have a cleft. / There is no direct transmission of nerve impulses. (1)
- Neurotransmitters are stored in vesicles in the endings of axons. (1)
- The arrival of a nerve impulse at the ending of an axon stimulates the release of neurotransmitters. (1)
- Neurotransmitters diffuse across the cleft. (1)
- Neurotransmitters bind to receptor sites on the postsynaptic membrane. (1)

Differences (Max. 4)

Synapse	Neuromuscular junction
• A junction between two neurones (1)	• A junction between a motor neurone and a muscle fibre (1)
• Neurotransmitters are released from vesicles in the synaptic knob (1)	• Neurotransmitters are released from vesicles in the motor end plate (1)
• Binding of neurotransmitters stimulates the postsynaptic neurone to generate a nerve impulse and continue with the transmission (1)	• Binding of neurotransmitters stimulates the membrane of the muscle fibre to generate an electrical impulse to cause the muscle to contract (1)

Communication (Max. 3)

(11 marks)

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Skills assessed	Marks
Understanding basic concepts	8
Communication	3

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